

Luís Carlos Fagundes

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Physicist · Nonequilibrium statistical mechanics

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EDUCATION

- 2022 – 2026
PhD in Physics
Universidade Federal do Rio Grande do Sul
Thesis: *Opinion models, inertia and coarsening*. Supervisor: Jeferson J. Arenzon. Co-supervisor: Wellington G. Dantas.
- 2020 – 2022
MSc in Physics
Universidade Federal do Rio Grande do Sul
Dissertation: *O papel de zelotes emergentes no modelo do votante*. Supervisor: Jeferson J. Arenzon. Co-supervisor: Wellington G. Dantas.
- 2012 – 2019
BSc in Physics
Universidade Federal do Rio Grande do Sul
Final thesis: *Estudo de modelos de rede por meio de algoritmos de execução em paralelo*. Advisor: Heitor C. M. Fernandes.

RESEARCH

- 2024–
Percolation in inter-cellular metabolic networks
With Andrea De Martino and Daniele De Martino. Percolation transitions of cross-feeding networks formed by cells in metabolic overflow.
- 2020–
Opinion inertia, zealots and coarsening
Instituto de Física, UFRGS. The persistent voter model: self-induced zealots, effective surface tension, interface geometry and finite-size scaling, and minimal random-walk models for the observed exponents.
- 2021
Synchronization of epidemic oscillations induced by social network control
Summer-school research, London Mathematical Laboratory. Supervisors: Daniele De Martino and Fabio Caccioli.

PUBLICATIONS

- 2025
Cross-feeding percolation phase transitions of inter-cellular metabolic networks
Latoski, L. C. F., De Martino, A., De Martino, D.
Science Advances **11**, eadv8216 (2025). doi:10.1126/sciadv.adv8216 arXiv:2412.09088
- 2024
Opinion inertia and coarsening in the persistent voter model
Latoski, L. C. F., Dantas, W. G., Arenzon, J. J.
Physical Review E **109**, 054115 (2024). doi:10.1103/PhysRevE.109.054115 arXiv:2403.02483
- 2022
Curvature-driven growth and interfacial noise in the voter model with self-induced zealots
Latoski, L. C. F., Dantas, W. G., Arenzon, J. J.
Physical Review E **106**, 014121 (2022). doi:10.1103/PhysRevE.106.014121 arXiv:2204.03402

PRESENTATIONS

- Feb 2026
Cross-feeding percolation phase transition of intercellular metabolic networks
2026 [simbad] School & Workshop, University of La Habana, Havana, Cuba. Oral.
- Sep 2025
Cross-feeding percolation phase transitions of inter-cellular metabolic networks
First Simbad Workshop, Cascina Fossata, Turin, Italy. Oral.

Apr 2025	São Paulo School of Advanced Science on Disordered Systems (SPSASDS) ICTP-SAIFR / IFT-UNESP, São Paulo, Brazil. Poster.
Nov 2024	Opinion inertia and coarsening in the persistent voter model Brazilian Meeting on Statistical Physics, Curitiba, Brazil. Oral.
May 2024	Opinion inertia and coarsening in the persistent voter model Encontro de Outono da Sociedade Brasileira de Física, Florianópolis, Brazil. Oral.
2024	Percolation of single-cell cross-feeding networks during metabolic overflow Encontro Nacional de Física Estatística, Brazil. With D. De Martino and A. De Martino.
Nov 2023	Learning probability distributions of neuronal activity patterns Minicourse on Lattice Models and Applications to Biological Problems, ICTP-SAIFR, São Paulo, Brazil.
May 2023	Curvature-driven growth and interfacial noise in the voter model with self-induced zealots Encontro de Outono da Sociedade Brasileira de Física, Ouro Preto, Brazil. Oral.
Jul 2021	Synchronization of epidemic oscillations induced by social network control London Mathematical Laboratory Summer School, online. Oral.
Nov 2021	Impact of self-induced disorder on the voter model Brazilian Meeting on Statistical Physics, online. Poster.
2019	Modelos de rede: uma abordagem com algoritmos de execução em paralelo XXXI Salão de Iniciação Científica, UFRGS, Porto Alegre, Brazil. Poster and oral.

SCHOOLS ATTENDED

Oct 2023	Advanced School on Quantitative Principles in Microbial Physiology: from Single Cells to Cell Communities ICTP, Trieste, Italy.
Oct 2021	Social Phenomena from a Physics Perspective ICTP-SAIFR, online.
Sep 2021	Statistical Physics of Complex Systems ICTP and EPS Statistical and Nonlinear Physics Division, online.

FELLOWSHIPS AND AWARDS

2022 – 2026	Doctoral fellowship, CNPq Full-time.
2020 – 2022	Master's fellowship, CAPES Full-time.
2019 – 2020	Scientific-initiation fellowship, UFRGS Advisor: Heitor C. M. Fernandes.
2019	Destaque, Interdisciplinary Computational Physics session — oral and poster XXXI Salão de Iniciação Científica, UFRGS.

TEACHING AND OUTREACH

	Independent tutor in physics Private tuition, one to one.
2015	Extension fellowship, UFRGS Interdisciplinary activities with primary-school pupils, supervised by Cláudia H. D. S. Jaskulski and Patrícia M. Vieira.

SKILLS

Languages	Portuguese (native), English (spoken and written)
Computing	C, C++, Python, OpenMP, LaTeX, Git Kinetic and stationary Monte Carlo simulation, finite-size scaling analysis, fractal-dimension estimators, master equations and mean-field closures.